Preliminary formal prior-art confirmation for ECR-MF

Scope and conclusion

This is a **preliminary patent-and-literature prior-art assessment**, not a legal opinion, claim-construction opinion, patentability opinion, or freedom-to-operate clearance. The searched concept was ECR-MF: an HEO-specific controller that switches among semi-analytical, regularized, and high-fidelity numerical propagation using predicted state, event-time, and covariance risk; performs local dual propagation at critical events; escalates fidelity when the risk budget is exceeded; and emits a machine-checkable certificate.

The search found **substantial partial prior art for the ingredients**, but not, in the retrieved records, a single reference that discloses the full combination as narrowly framed for highly eccentric Earth orbits. The strongest prior-art risk is therefore not that "adaptive propagation" is new; it is that broad claims covering multifidelity orbit propagation, model selection, error correction, uncertainty ensembles, or event-aware switching may read on existing work.

The defensible position is: **ECR-MF may have a patentable or publishable distinction as a specific HEO event-risk/covariance-aware switching and certification policy, but novelty is not confirmed until the claims are narrowed and a professional search covers claims, continuations, family members, non-patent literature, and relevant CPC classes.**

Closest retrieved prior art

1. Non-averaged regularized propagation for HEO

Amato et al. compare semi-analytical and non-averaged regularized propagation, decompose integration error into dynamical, model-truncation, short-periodic, and numerical components, and test regularized methods on LEO, GTO, and high-altitude HEO. They report that regularized non-averaged methods can be substantially faster than semi-analytical methods for the tested HEO case.[1]

ECR-MF implication: regularized/non-averaged propagation and the HEO use case are not new. Do not claim those elements alone.

2. Adaptive multifidelity uncertainty propagation

Jones and Weisman use low-fidelity propagation to construct a surrogate, selectively re-propagate important samples with a high-fidelity model, and correct the surrogate for orbit uncertainty propagation.[2] Related multifidelity orbit-determination work uses a hierarchy of force models and propagation methods and reports a twelve-fold filter-runtime reduction in the tested case without statistically significant filter-performance change.[3]

ECR-MF implication: broad claims for low/high-fidelity switching, selective high-fidelity samples, uncertainty-aware acceleration, or orbit-determination runtime reduction are high-risk.

3. Multisegment propagation for highly elliptical orbits

A modified Chebyshev–Picard approach uses nonuniform Chebyshev–Gauss–Lobatto sampling and multiple true-anomaly-based segments to obtain accurate solutions across eccentric orbits.[4]

ECR-MF implication: segmentation by orbital phase is not new. A claim should require the particular event-risk and covariance-driven switching rule, not merely true-anomaly segmentation.

4. Verified interval orbit propagation

Verified regularized interval propagation provides mathematically guaranteed position and velocity solutions over time and is positioned for conjunction analysis and related applications.[5]

ECR-MF implication: machine-verifiable enclosures are not new in isolation. The potential distinction is selective use of verification at HEO event-risk boundaries, tied to a propagation-mode decision and covariance criterion.

5. **Patent: multi-model ensemble atmospheric-density propagation**

US20240037297A1 discloses multiple trained density-prediction models, orbit propagation using the respective density outputs, and ensemble analysis of predicted orbit states to quantify uncertainty and determine an orbit state.[6]

ECR-MF implication: claims covering multiple environmental models, ensemble orbit propagation, or uncertainty-aware state prediction are exposed. The retrieved disclosure is focused on atmospheric-density model ensembles rather than HEO event-certified switching among propagation formulations.

6. **Patent: physics-plus-machine-learning trajectory correction**

US12046129B2 discloses an iterative physics trajectory prediction refined by a machine-learning trajectory-error model, with a pseudo-measurement and pseudo-measurement noise used to update the future trajectory.[7]

ECR-MF implication: claims covering a physics baseline plus learned error correction, pseudo-measurements, or model-error-based trajectory updates are exposed. ECR-MF should not require machine learning unless the specific trained error model is itself the intended contribution.

7. **Patent: transformer-based orbital prediction**

WO2025065111A1 discloses transformer-based resident-space-object trajectory prediction using state vectors and Keplerian data, including a forecast horizon and recursive use of predictions.[8]

ECR-MF implication: generic claims to AI-based orbital prediction are exposed and should be avoided. This reference does not, from the retrieved summary, disclose HEO event-risk switching, high-fidelity certification, or covariance-gated mode selection.

8. **Patent: distributed local orbit propagation**

US7612712B2 discloses a server/client architecture in which a server distributes seed state vectors and propagation-model coefficients so a client can locally propagate satellite orbits.[9]

ECR-MF implication: distributed deployment and local propagation are not new. They should be treated as implementation context, not the inventive core.

Element-by-element assessment

Proposed ECR-MF element	Prior-art exposure	Assessment
Semi-analytical, regularized, and high-fidelity propagation modes	High	Known individually and in combinations; not a defensible standalone novelty point.
Switching between low- and high-fidelity propagation	High	Multifidelity orbit uncertainty and OD literature directly covers selective high-fidelity correction.[2, 3]
HEO-specific true-anomaly or event-based segmentation	Medium–high	Multisegment HEO propagation is known.[4]
Predicted event-time error as a switching variable	Medium	The search found evidence that event-time error is a real regularization issue, but not a single retrieved patent disclosing the proposed ECR-MF decision policy.[10]

Proposed ECR-MF element	Prior-art exposure	Assessment
Covariance consistency as a switching variable	Medium	Multifidelity uncertainty and OD methods are known; the specific joint rule with HEO event-time risk was not identified in the retrieved set.[2, 3]
Short dual propagation at perigee, burns, eclipse, and covariance-risk boundaries	Medium–low in the retrieved set	This is the most promising differentiator, but the search does not prove absence of closer patent or paper disclosures.
Rollback to the last certified state when disagreement exceeds a local budget	Medium–low in the retrieved set	Potentially differentiating as a control policy; must be claimed with precise state, event, and certification conditions.
Machine-checkable certificate containing mode, force model, event bound, covariance status, and switch reason	Medium–low in the retrieved set	Potentially useful as a system/method limitation, but "certificate" alone is unlikely to establish novelty.
Selective verified interval propagation only at critical HEO segments	Medium	Verified interval propagation exists; the selective coupling to event-risk switching may be the differentiating combination.[5]

Recommended claim strategy

Avoid an independent claim framed as "an adaptive multifidelity orbit propagator." That is too broad and directly exposed by the multifidelity literature and related patents.

A stronger independent method claim should require all of the following as a linked sequence:

1. receiving a highly eccentric Earth-orbit state, force-model configuration, covariance, and mission event thresholds;
2. propagating a first segment using a selected low-cost or regularized method;
3. predicting both state error and event-time/covariance risk for a future perigee, maneuver, eclipse, apogee, or ground-contact event;
4. initiating a second, independent certification propagation only when the predicted risk reaches a defined boundary;
5. comparing state, event-time, and covariance outputs from the two methods;
6. accepting the low-cost propagation only when all three error conditions remain within budget, otherwise rolling back or escalating fidelity; and
7. emitting a machine-readable certificate recording the selected mode, force model, event bound, covariance test, and escalation decision.

The claim should specify how the event-time bound is computed, how covariance inconsistency is detected, what triggers rollback, and how the certified state is transferred between formulations. Those details—not the labels "adaptive," "multifidelity," "regularized," or "AI"—are where technical distinction is most likely to reside.

Confirmation status

Preliminary confirmation: partial prior art found; full-combination anticipation not found in the retrieved set. The ECR-MF concept is not cleanly clear of prior art. Its broad ingredients are substantially occupied, while the narrower combination of HEO event-risk prediction, covariance-aware mode switching, local dual certification, rollback, and machine-readable proof remains a plausible research and patent angle.

Before filing or making a definitive novelty statement, perform:

- a professional patent search of US, WO, EP, CN, JP, and KR families;
- family and continuation analysis for the closest patents;
- full claim-chart review, not abstract-only review;
- CPC/IPC searches for orbit determination, space-object tracking, uncertainty propagation, numerical integration, and spacecraft guidance;
- non-patent literature search through the earliest priority dates;
- an attorney-led novelty, inventive-step, and freedom-to-operate assessment.

References

1. Amato D, Bombardelli C, Baù G, et al (2018) Non-averaged regularized formulations as an alternative to semi-analytical orbit propagation methods. Celestial mechanics & dynamical astronomy. https://doi.org/10.1007/s10569-019-9897-1

2. Jones B, Weisman R (2019) Multi-fidelity orbit uncertainty propagation. Acta Astronautica. https://doi.org/10.1016/J.ACTAASTRO.2018.10.023

3. Jones B (2018) Multi-Fidelity Methods for Orbit Determination

4. Kim D, Junkins J, Turner J (2015) Multisegment Scheme Applications to Modified Chebyshev Picard Iteration Method for Highly Elliptical Orbits. Mathematical Problems in Engineering. https://doi.org/10.1155/2015/290781

5. Geul J, Mooij E, Noomen R (2021) Verified Regularized Interval Orbit Propagation. Journal of Guidance Control and Dynamics. https://doi.org/10.2514/1.G004947

6. System to Determine the State of an Orbiting Object Using Multi-Model Ensemble Techniques for Drag and Methods Thereof. https://patents.google.com/patent/US20240037297A1/en

7. Computer-based systems configured for space object orbital trajectory predictions and methods thereof. https://patents.google.com/patent/US12046129B2/en

8. Transformer neural network systems and methods for enhanced orbital prediction accuracy. https://patents.google.com/patent/WO2025065111A1/en

9. Distributed orbit modeling and propagation method for a predicted and real-time assisted GPS system. https://patents.google.com/patent/US7612712B2/en

10. Urrutxua H, Álvarez J, Bombardelli C (2016) Impact of the integration stop-condition on the accuracy of regularized orbit formulations